

Filter-matched rate calibration for variable-rate PING training

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Abstract

Selecting a Poisson input rate for conductance-based spiking networks requires accounting for the filtering and variability introduced by synapses, membrane integration, and finite observation windows. We propagated MNIST pixel spike trains through the target AMPA and subthreshold membrane dynamics, characterized their empirical responses, and derived a corresponding linear filter model. Nonlinear and linear decoders were then trained on fresh direct simulations across the registered rate grid. Held-out nonlinear-decoder accuracy was reliably above chance at 0.25 Hz and reliably exceeded 50% from 0.5 Hz. This practical floor was unchanged across 0.6, 1.2, and 2.4 μ S probes, supporting 0.5–25 Hz for subsequent variable-rate PING training.

Purpose and scope

The question is:

At a given encoding rate, does the temporally filtered image still contain enough information for an ANN to recognize the digit?

The target PING architecture has 784 pixel channels feeding conductance-based excitatory cells. Its fixed synaptic and membrane equations define the features, without trained input, recurrent, or output weights. The ANN thus measures filtered-input decodability independently of a trained PING network.

We characterized these features empirically and analytically, trained matched nonlinear and linear decoders on fresh simulations, and evaluated frozen models on held-out images. The resulting thresholds measure decoder-relative accessibility, not an absolute information boundary or PING accuracy.

Methods

1. **Generated and validated filter-matched pixel features.** Every image used a 200 ms presentation and 0.1 ms timestep. The tested rates were 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz, sampled densely below 5 Hz and capped at 25 Hz, a range previously effective in trained PING networks. Seeds 42, 43, 44 defined independent feature and empirical-response runs.

The rate grid and deterministic MNIST training and validation indices were recorded before analysis; the official MNIST test set was held out and not loaded. Pixel intensities x

lay in $[0, 1]$. If r was the encoding rate at pixel intensity one, the expected pixel rate was $r \cdot x$.

Each pixel drove an independent AMPA synapse and uncoupled non-spiking excitatory membrane. This captured synaptic decay, conductance-dependent integration, and within-window timing, but included no threshold, reset, recurrence, or trained weights.

The complete feature path is

$$S_i(t) \rightarrow g_i^{\text{pix}}(t) \rightarrow v_i(t) \rightarrow z_i \rightarrow \text{ANN}. \quad (1)$$

Here S_i , g_i^{pix} , v_i , and z_i are pixel i 's spike train, AMPA conductance, subthreshold voltage, and time-averaged feature; t is time and ANN is the classifier.

At each timestep, we drew the encoded input according to

$$S_i(t) \sim \text{Bernoulli}(r \Delta t x_i). \quad (2)$$

Here x_i is grayscale pixel-intensity, r is the encoding rate in spikes per second at pixel-intensity one, Δt is the timestep in seconds, and Bernoulli is an independent binary draw.

We updated the AMPA conductance using

$$g_i^{\text{pix}}(t) = \beta_{\text{AMPA}} g_i^{\text{pix}}(t-1) + w_{\text{probe}} S_i(t). \quad (3)$$

$$\beta_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (4)$$

Here β_{AMPA} is one-timestep conductance retention, τ_{AMPA} is AMPA decay time, w_{probe} is conductance added per spike, and $\exp$ is the exponential function. Other symbols follow Equation 2. The primary $1.2 \mu\text{S}$ probe is the target architecture's nominal mean initial pixel-to-excitatory conductance; 0.6 and $2.4 \mu\text{S}$ test half and double scale for comparison.

We integrated the membrane without thresholding or resetting:

$$C_E \frac{dv_i}{dt} = g_{L,E}(E_L - v_i) + g_i^{\text{pix}}(t)(E_e - v_i). \quad (5)$$

$$\tau_{\text{eff},i}(t) = \frac{C_E}{g_{L,E} + g_i^{\text{pix}}(t)}. \quad (6)$$

Here C_E is excitatory capacitance; $g_{L,E}$ is leak conductance; E_L and E_e are leak and AMPA reversal potentials; and $\tau_{\text{eff},i}$ is the instantaneous conductance-dependent time constant. Other symbols follow Equation 1.

Voltage began at E_L and conductance at zero. After one presentation,

$$z_i = \frac{1}{T} \int_0^T (v_i(t) - E_L) dt. \quad (7)$$

where T is presentation duration and z_i is mean baseline-subtracted voltage. Neither the feature nor the decoder input included rate, so sparse inputs retained their weaker, noisier signal.

2. **Characterized the empirical response moments.** For visualization and consistency checks, we used 512 independent simulations per rate–intensity–conductance condition and seed. This was a practical diagnostic sample size, not a convergence gate or a requirement for ANN training. The empirical response table estimated the conditional mean

$$\hat{\mu}_z(x, r, w_{\text{probe}}) = \frac{1}{K} \sum_{k=1}^K z^k. \quad (8)$$

and conditional variance

$$\hat{\sigma}_z^2(x, r, w_{\text{probe}}) = \frac{1}{K-1} \sum_{k=1}^K (z^k - \hat{\mu}_z)^2. \quad (9)$$

Here x , r , and w_{probe} are pixel-intensity, rate, and probe conductance; $z^{(k)}$ is draw k ; K is the draw count; and Equations 8 and 9 estimate the conditional mean and variance.

The table served only to plot moments and check consistency. Decoder inputs instead used a fresh spike train and direct synapse–membrane simulation on every presentation n :

$$S_i^n(t) \rightarrow g_i^{\text{pix},n(t)} \rightarrow v_i^n(t) \rightarrow z_i^n \rightarrow \text{ANN}. \quad (10)$$

Repetition therefore sampled the full conditional distribution of z_i without an intermediate noise model.

Exact replay, finite values, physical bounds, zero-intensity resting response, independent streams, recomputed moments, fresh direct simulations, and float32 fidelity were checked.

3. **Derived the analytical linear-filter response.** We calculated a local linear approximation to the synapse, membrane, and finite averaging window. This diagnostic was not used to select a training range.

For mean spike rate λ , stationary mean conductance is

$$|g|_\lambda = \lambda w_{\text{probe}} \tau_{\text{AMPA}}. \quad (11)$$

and stationary mean voltage is

$$|v|_\lambda = \frac{g_{\text{L,E}} E_L + |g|_\lambda E_e}{g_{\text{L,E}} + |g|_\lambda}. \quad (12)$$

Here λ is encoding rate times pixel-intensity; $\text{bar}(g)_\lambda$ and $\text{bar}(v)_\lambda$ are stationary mean conductance and voltage. Other symbols follow Equations 3–7.

Linearizing the synapse and membrane around that operating point gives

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\text{L,E}} + |g|_\lambda}. \quad (13)$$

Here $G_\lambda(\omega)$ is the local synapse-plus-membrane transfer function; ω is angular frequency; and i is the imaginary unit. Because mean conductance changes gain and effective membrane time constant, Equation 13 defines a family of responses.

The finite averaging window contributes

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}. \quad (14)$$

$$H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (15)$$

Here $A_T(\omega)$ is the duration- T averaging response; $H_\lambda(\omega)$ is the complete response from input fluctuation to averaged feature; T is presentation duration; and $\exp$ is the exponential function.

For centred ideal Poisson input,

$$S_{\text{in}(\omega)} = \lambda. \quad (16)$$

$$S_{z(\omega)} = |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)}. \quad (17)$$

$$\text{Var}_{\text{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)} d\omega. \quad (18)$$

Here $S_{\text{in}}(\omega)$ and $S_z(\omega)$ are input and predicted output power spectral densities; $|H_\lambda(\omega)|^2$ is transmitted noise power; $\text{Var}_{\text{linear}}(z)$ is predicted feature variance; and π is the circle constant. Appendix A derives Equations 11–18.

We evaluated the transfer functions at every distinct calibration point and validated low-, middle-, and high-drive gains by sinusoidally modulating the numerical probe. This stationary, continuous-time, linearized Poisson model approximates the finite, discrete Bernoulli simulation and remains diagnostic only.

Without new simulation, we compared these predictions with the Step 2 table at every rate, intensity, and conductance. Analytical mean was Equation 12 minus resting voltage; analytical SD was the square root of Equation 18. Predicted-versus-empirical plots used an identity line, Pearson correlation, mean absolute error, and median prediction/empirical ratio. Summary statistics excluded conditions where both responses were zero.

4. **Constructed and compared complete feature images.** We used only the official 60,000-image MNIST training partition. Indices 0–54,999 and 55,000–59,999 were reserved for decoder training and validation, respectively; the official MNIST test set was held out and not loaded. For each uint8 pixel, the diagnostic sampler used its exact 0–255 intensity index and selected an authenticated empirical draw with independent, deterministic pixel and image streams. A fresh direct simulation provided the comparison.

The comparison covered training images 0–15, all 28×28 pixels, eight replicates, three conductances, and 0.25, 3, and 25 Hz. Tests covered pooled and image-level moments, zero fractions, absolute and relative differences, and spatial correlation of per-pixel means. The low-rate image-level checks failed; this limitation was retained, while subsequent decoders used fresh direct simulation rather than the table.

5. **Trained mixed-rate decoders.** Each simulated image produced a feature vector $\mathbf{z} \in \mathbb{R}^{784}$, containing one time-averaged membrane-voltage displacement per pixel. The primary ANN transformed this vector through a fully connected 1,024-unit hidden layer,

$$\mathbf{h} = \text{ReLU}(W_h \mathbf{z} + \mathbf{b}_h), \quad \mathbf{s} = W_o \mathbf{h} + \mathbf{b}_o,$$

where $\mathbf{h}$ is hidden activation, $\mathbf{s} \in \mathbb{R}^{10}$ contains one logit per digit, and elementwise $\text{ReLU}(x) = \max(0, x)$ retains positive inputs and zeros negative ones. Both weight matrices and bias vectors were trainable; prediction used $\arg \max_c s_c$. The 784–1,024–10 network had no convolution, recurrence, normalization, dropout, or pretrained weights.

The matched linear decoder omitted the hidden layer and computed

$$\mathbf{s} = W_{\text{lin}} \mathbf{z} + \mathbf{b}_{\text{lin}}.$$

Its 784×10 weights and ten biases were trainable. “Linear” described the feature-to-logit map, not fitting: both models used backpropagation. Linear performance measured directly accessible digit evidence; the ANN advantage measured the benefit of its nonlinear representation. Neither received rate.

Both models were fitted with ten-class cross-entropy. For logits $\mathbf{s}$ and true digit label y , the loss for one image was

$$\mathcal{L}_{\text{CE}(\mathbf{s}, y)} = -\log \left(\frac{\exp(s_y)}{\sum_{c=0}^9 \exp(s_c)} \right).$$

This penalized low probability for the correct digit; backpropagation trained each decoder’s weights and biases.

Each presentation sampled the registered rates uniformly, then generated z_i from a fresh spike train through Equations 3–7. Both decoders saw the same feature batch. Adam used learning rate 0.001, batch size 256, and 15 epochs. The ANN used no weight decay; three linear candidates used L2 weight decays 10^{-5} , 10^{-4} , and 10^{-3} validation accuracy selected the decay and, independently for each decoder, the retained epoch. Seeds 42, 43, 44 initialized parameters and sampling streams. The MNIST test set remained held out through fitting and selection. The primary 1.2 μS ensemble and separate 0.6 and 2.4 μS ensembles were not pooled.

6. **Evaluated frozen held-out psychometric curves.** All decoders, selected epochs, hyperparameters, checkpoint hashes, rates, direct-simulation seeds, and uncertainty rules were committed before the held-out MNIST test set was accessed. At each rate, every decoder seed received the same 10,000 held-out images and three reproducible fresh direct-simulation draws. The primary curve was

$$A_r(r) = P(\text{correct} \mid r, \text{mixed-rate nonlinear decoder}). \quad (19)$$

Here A_r is held-out nonlinear-ANN accuracy, P is correct-classification probability, and r is rate. The linear decoder was diagnostic.

Two thousand bootstrap repetitions resampled images, simulation draws, and decoder seeds to give the one-sided 95% lower bound L_r . The decoder-relative edge was

$$r_{\text{decode}} = \text{lowest } r \in \mathcal{R} \text{ satisfying } L_r(r) > \frac{1}{N_{\text{class}}}. \quad (20)$$

The practical training floor was

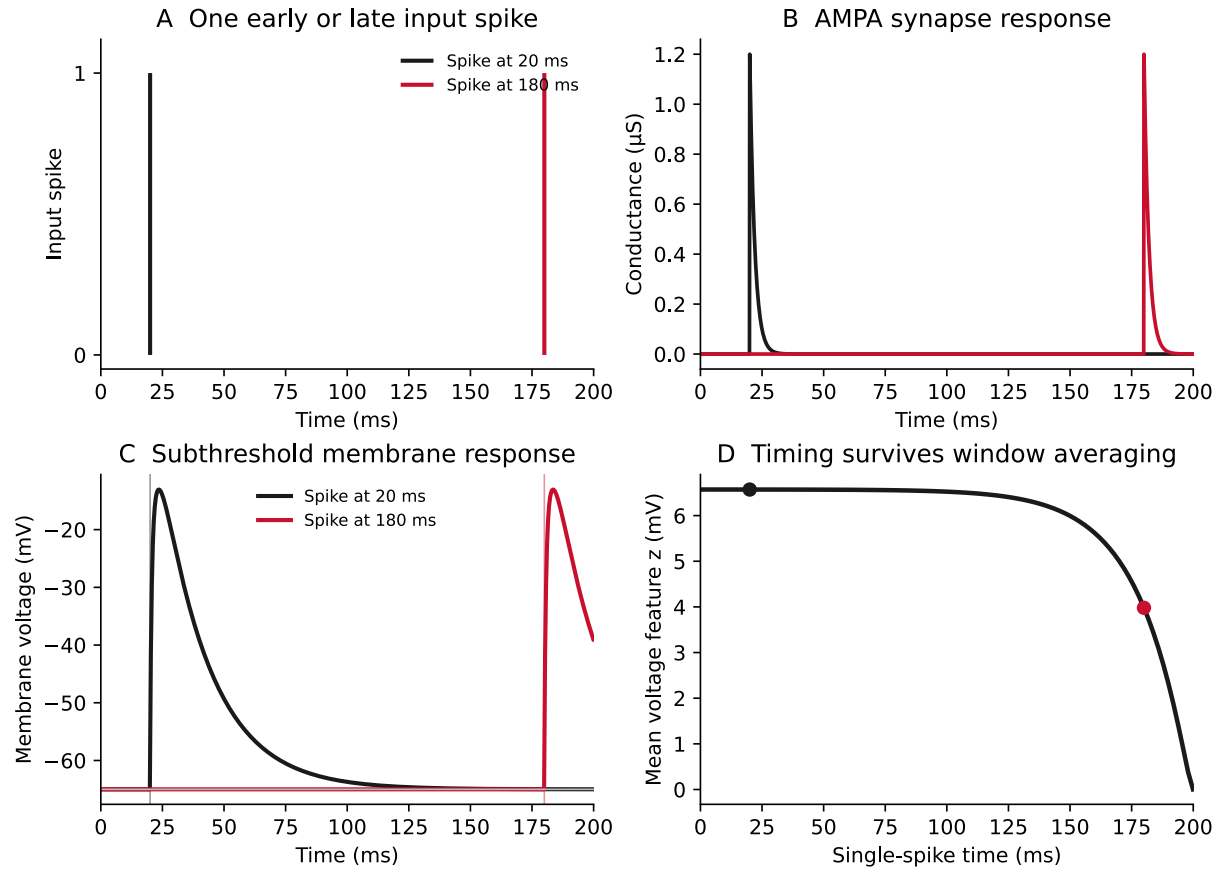
$$r_{\text{train}} = \text{lowest } r \in \mathcal{R} \text{ satisfying } L_r(r) \geq a_{\text{use}}. \quad (21)$$

Here $\mathcal{R}$ is the tested grid, N_{class} the class count, r_{decode} the lowest rate reliably above chance, a_{use} the 50% useful-accuracy target, and r_{train} its first reliable crossing. Primary thresholds used the 1.2 μS ANN; the linear decoder and other conductances measured sensitivity. Interpolated rates were not observations.

7. **Selected the training range.** The decision reported r_{decode} , r_{train} , uncertainty, conductance sensitivity, model and artifact hashes, and every registered rule outcome. The later PING range began at r_{train} and ended at the registered 25 Hz ceiling. The rule would have reported a range of plausible floors if conductance shifted the practical floor by more than one adjacent grid point.

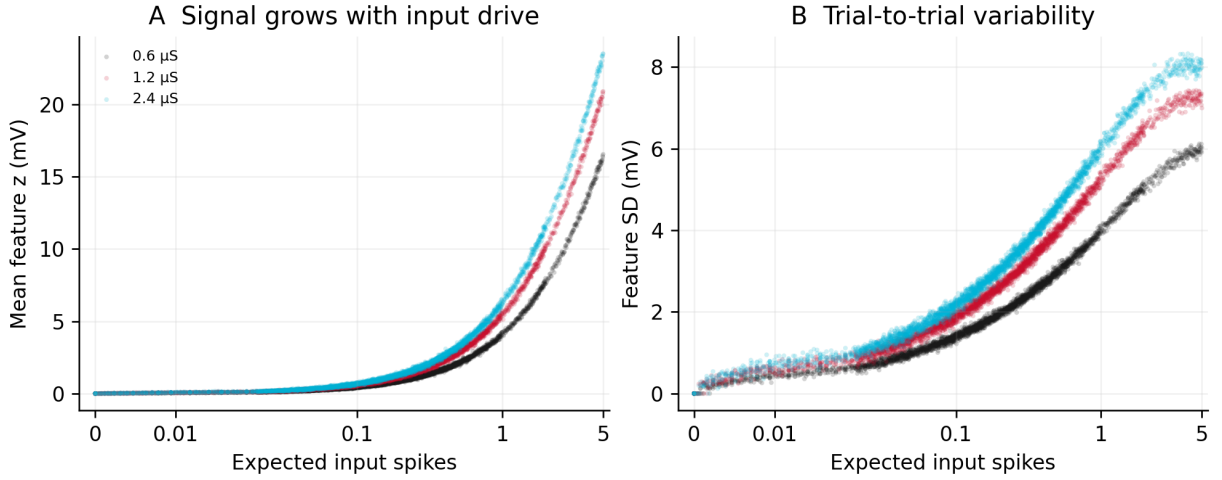
Results

Filter-matched feature generation



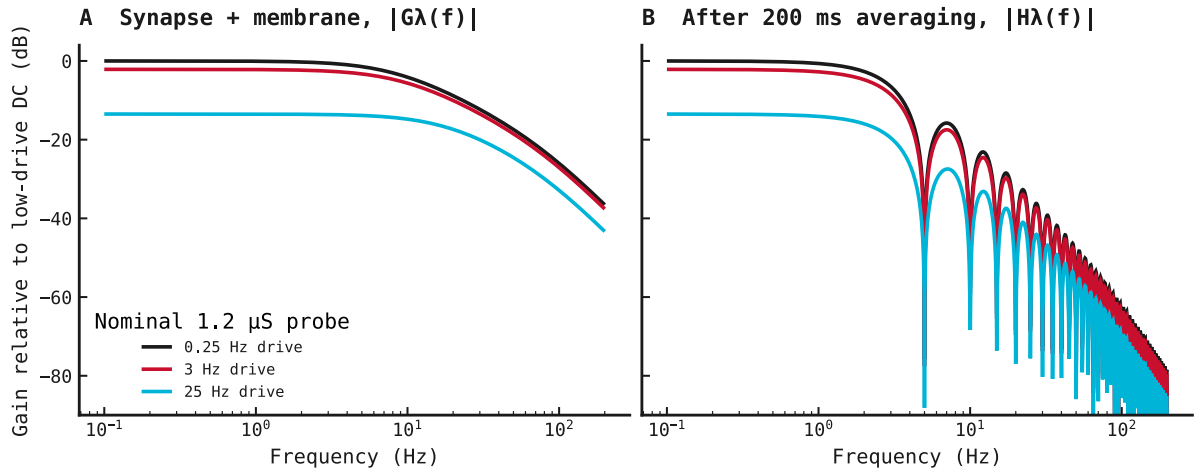
Finite-window timing sensitivity. Panel A places one input spike at 20 or 180 ms. Panels B and C pass those inputs through the registered AMPA synapse and subthreshold membrane. Panel D repeats the simulation across 101 single-spike times and plots the presentation-averaged feature z . Conductance and voltage rise after each spike and then relax through their respective decay dynamics; z falls for late spikes because the fixed 200 ms window truncates more of their response.

Empirical response table



Signal and variability across the empirical response table. For every rate–intensity condition, Panel A plots the mean feature z and Panel B its standard deviation from 512 independent simulations per condition and seed, against encoding rate $\times$ normalized pixel intensity $\times$ 200 ms. Each point is one rate–intensity condition, and colour encodes probe conductance. Both moments rise because larger expected spike counts deliver more stochastic conductance; larger probes produce larger voltage excursions.

Analytical linear-filter response



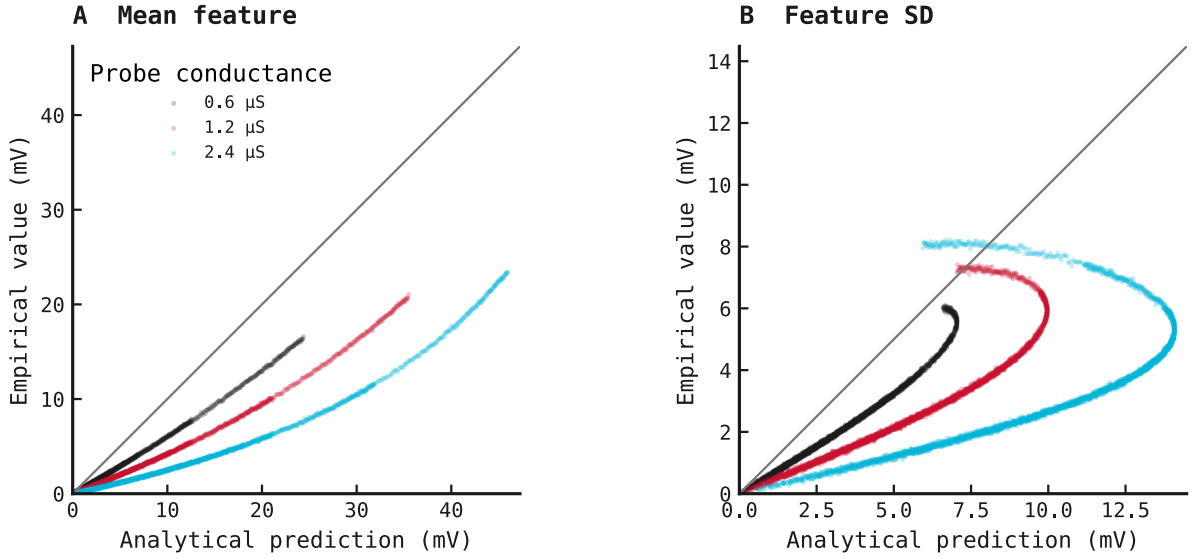
Analytically calculated linearized frequency response at the nominal 1.2 μS probe. These analytical functions take modulation frequency, mean input rate, and probe conductance; they contain no simulated traces. Frequency is a small sinusoidal rate modulation around 0.25, 3, or 25 Hz. Panel A evaluates the magnitude of Equation 13 from 0.1–200 Hz, with $\omega = 2\pi f$. Gain is

$$20 \log_{10} \left(\frac{|G_{\lambda(2\pi f)}|}{|G_{\lambda_{\text{low}}}(0)|} \right),$$

relative to the 0.25 Hz DC response. Thus the black curve begins near 0 dB; higher mean conductance lowers the red and cyan DC gains. Panel B plots Equation 15, $H_{\lambda} = A_T G_{\lambda}$, after the analytical 200 ms rectangular-window average.

Panel A combines an AMPA low-pass pole at $\frac{1}{\tau_{\text{AMPA}}}$ with a membrane pole set by effective conductance and capacitance, smoothly suppressing fast modulation. Greater drive both shunts the membrane and depolarizes it toward E_e , reducing excitatory driving force and separating the curves.

Panel B adds the averaging magnitude $|A_{T(2\pi f)}| = \left| \frac{\sin(\pi f T)}{\pi f T} \right|$. It is one at zero frequency and zero at $f = \frac{n}{T}$ for nonzero integer n . With $T = \frac{200}{1000}$ s, integer cycles cancel at 5, 10, 15 Hz, and so on. Magnitude folds alternating sinc signs into positive lobes; the two low-pass terms attenuate successive lobes to form the falling envelope.



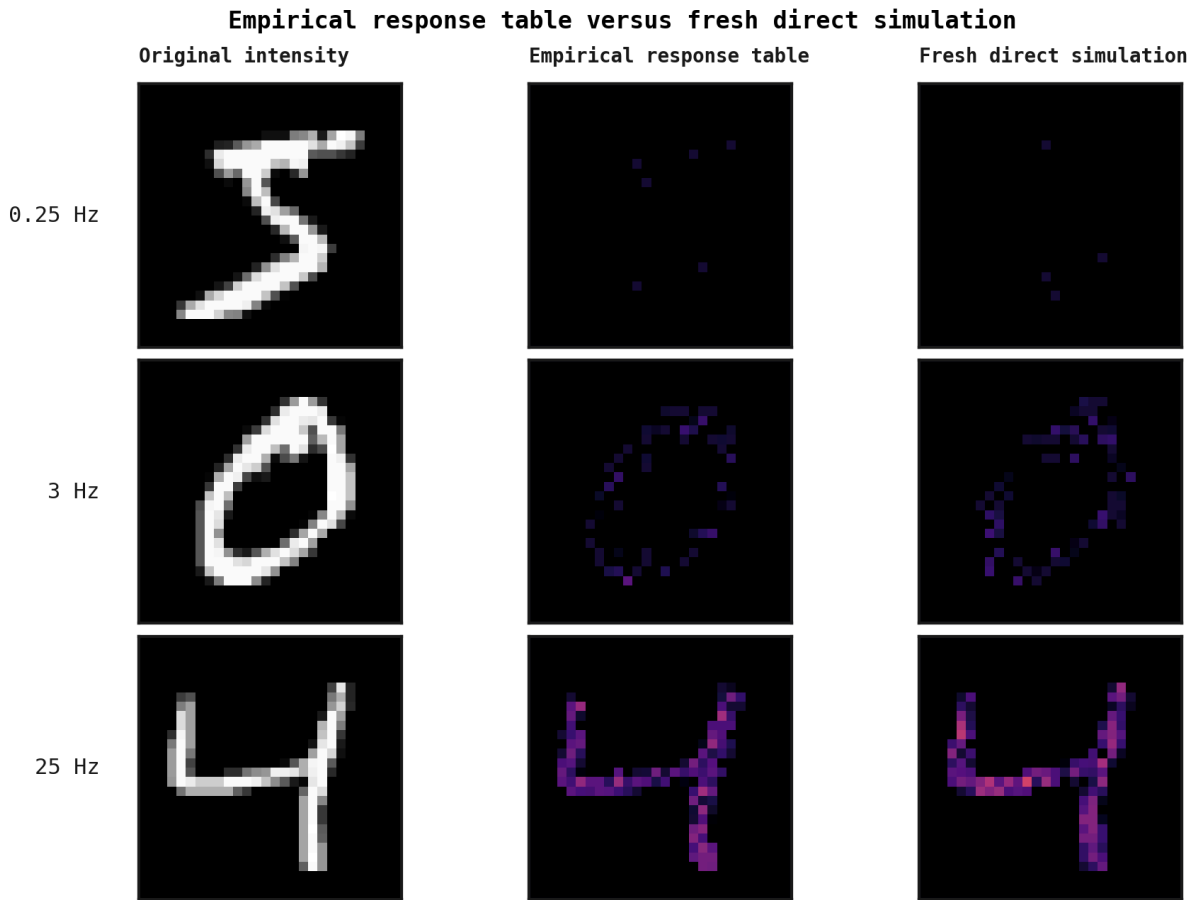
Stationary analytical predictions versus the empirical response table. Each point is one recorded rate–intensity condition; colour denotes probe conductance. Panel A compares the stationary mean displacement from Equation 12 with the empirical 200 ms feature mean. Panel B compares the square root of the Equation 18 variance with empirical feature SD. The diagonal denotes exact agreement. The analytical model preserved the broad ordering of the mean responses (Pearson $r = 0.951$) but predicted a median 2.52-fold larger mean. SD agreement was weaker ($r = 0.741$), with a median 2.245-fold overprediction. Points therefore lie mainly below the identity line because the stationary, locally linear model omits the start-from-rest transient, finite-window nonstationarity, discrete Bernoulli input, and nonlinear conductance fluctuations present in the empirical simulations.

Panel A bends upward because Equation 12 evaluates voltage at stationary mean conductance, whereas empirical trajectories begin at rest and average only 200 ms. Sparse presentations often contain no event or a late one; with greater drive, earlier and more consistent events let the empirical mean approach stationarity. Probe curves separate because a mean conductance can comprise many small events or fewer large ones. Larger jumps amplify fluctuations, and voltage saturation gives approximately $E[v(g)] < v(E[g])$.

Panel B hooks because variability rises from zero as spike counts and timings diversify, then falls when averaging suppresses relative count fluctuations and shunting and saturation limit voltage excursions. Equation 18 likewise balances rising Poisson noise against falling local

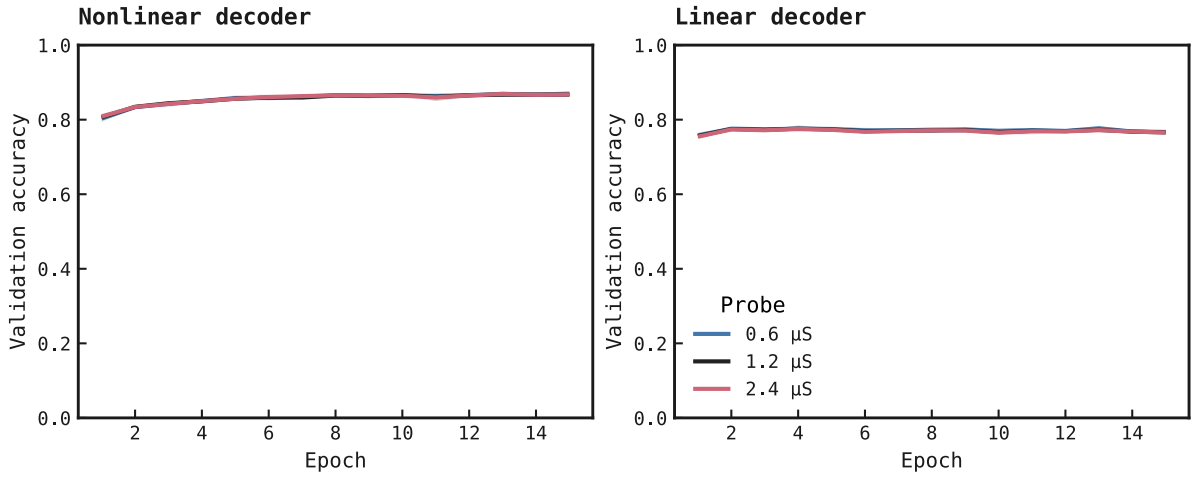
gain, but places the maximum elsewhere. Plotting these displaced maxima against each other creates the arches and backward branches; larger, less linear probe events enlarge them.

Complete feature images



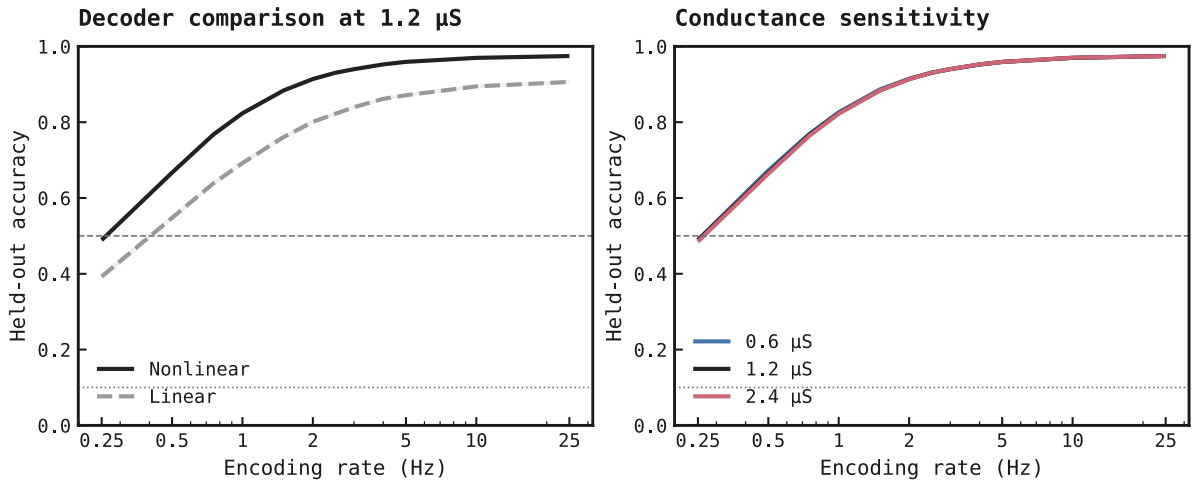
Empirical response table samples versus fresh direct simulations. Rows show one original MNIST image, one authenticated empirical response table sample, and one fresh direct simulation at 0.25, 3, and 25 Hz for the nominal 1.2 μ S probe, using common 0–65 mV limits. Low-rate images are sparse because most pixels receive no spike; increasing rate reveals the intensity pattern because spatial signal becomes larger relative to independent sampling noise.

Mixed-rate decoder training



Mixed-rate validation histories. Each panel plots mean validation accuracy across decoder seeds, with the seed range shaded, for models trained from fresh direct simulations at uniformly sampled registered rates. Curves rise rapidly and then flatten because most learnable class structure is acquired in the first several epochs; the three conductances overlap because scaling the subthreshold response preserves nearly the same spatial evidence. At the nominal probe, validation selected nonlinear epochs 10, 12, and 15, with accuracies 86.98%, 86.88%, and 86.94%. The matched linear decoders reached 78.36%, 77.34%, and 78%.

ANN psychometric curve and thresholds



Frozen held-out psychometrics. Curves show accuracy on the same 10,000 held-out MNIST images from three direct-simulation draws and three decoder seeds; bands show the registered hierarchical 95% interval. The dotted and dashed horizontal rules mark 10% chance and 50% useful accuracy. Accuracy rises with rate because more pixels receive spikes within 200 ms, while the near-overlap across probe conductances shows that the trained nonlinear decoder compensates for their response-scale difference.

At 0.25 Hz, nominal nonlinear accuracy was 48.89%, with a 48.22% one-sided lower bound. At 0.5 Hz it was 66.69%, with a 66.09% lower bound. The nominal linear decoder also first crossed the 50% lower-bound criterion at 0.5 Hz.

Training-range decision

The nonlinear decoder was reliably above chance from 0.25 Hz, and its practical floor was 0.5 Hz. The practical floor remained 0.5 Hz at 0.6 μ S, 0.5 Hz at 1.2 μ S, and 0.5 Hz at 2.4 μ S. We therefore recommend rates from 0.5 Hz to 25 Hz for later variable-rate PING training. This is a decoder-relative practical range, not an absolute information limit or a PING accuracy result.

Relation to prior work

Filtered conductance shot noise can violate fixed-time-constant models [1], and Gaussian approximations can miss skewed responses [2], supporting the empirical response table. Decoder performance is not absolute neural information [3], and linear and nonlinear decoders can differ [4], motivating a decoder-relative edge and both decoder diagnostics.

References

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2. Brigham & Destexhe: *Nonstationary Filtered Shot-Noise Processes and Applications to Neuronal Membranes*. Physical Review E, 2015. doi:10.1103/PhysRevE.91.062102
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4. Warland, Reinagel & Meister: *Decoding Visual Information From a Population of Retinal Ganglion Cells*. Journal of Neurophysiology, 1997. doi:10.1152/jn.1997.78.5.2336

Appendix A: linear-filter derivation

This appendix derives Equations 11–18 from the same AMPA synapse and subthreshold membrane used by the empirical probe. It replaces the finite Bernoulli drive with a continuous-time Poisson point process and linearizes the membrane around the mean drive at each calibration condition.

A.1 Stationary operating point

Represent pixel i 's idealized spike train and synaptic conductance by

$$s_{i(t)} = \sum_k \delta(t - t_k), \quad \frac{dg_{i(t)}}{dt} = -\frac{g_{i(t)}}{\tau_{\text{AMPA}}} + w_{\text{probe}} s_{i(t)}. \quad (\text{A1})$$

Here $s_i(t)$ is a sum of Dirac impulses; t_k is spike k 's time; δ is the Dirac delta; $g_i(t)$ is AMPA conductance; τ_{AMPA} is its decay time; w_{probe} is conductance added per spike; and t is time.

Equation A1 is the continuous-time counterpart of the decay-then-add update in Equation 3.

For stationary Poisson rate λ , the mean conductance satisfies

$$0 = -\frac{\bar{g}_\lambda}{\tau_{\text{AMPA}}} + w_{\text{probe}} \lambda, \quad \bar{g}_\lambda = \lambda w_{\text{probe}} \tau_{\text{AMPA}}. \quad (\text{A2})$$

Here λ is encoding rate times pixel-intensity, and $\bar{g}_\lambda$ is stationary mean conductance.

The zero on the left states that the mean no longer changes with time.

Setting the mean membrane derivative in Equation 5 to zero gives

$$0 = g_{L,E}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda), \quad |v|_\lambda = \frac{g_{L,E}E_L + |g|_\lambda E_e}{g_{L,E} + |g|_\lambda}. \quad (\text{A3})$$

Here $\bar{v}_\lambda$ is stationary mean voltage; $g_{L,E}$ is leak conductance; and E_L and E_e are leak and AMPA reversal potentials. Other symbols follow Equation A2.

A.2 Local synapse-plus-membrane response

Write each signal as its operating point plus a small fluctuation:

$$g_i = |g|_\lambda + \delta g_i, \quad v_i = |v|_\lambda + \delta v_i, \quad s_i = \lambda + \delta s_i. \quad (\text{A4})$$

Here δg_i , δv_i , and δs_i are complete conductance, voltage, and input perturbation variables around the means defined in Equations A2 and A3.

To derive Equation A5, substitute the conductance and input decompositions from Equation A4 into the synapse equation, Equation A1:

$$\frac{d(|g|_\lambda + \delta g_i)}{dt} = -\frac{|g|_\lambda + \delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}(\lambda + \delta s_i).$$

The operating point is stationary, so its time derivative is zero. Expanding the right-hand side then gives

$$\frac{d\delta g_i}{dt} = \left(-\frac{|g|_\lambda}{\tau_{\text{AMPA}}} + w_{\text{probe}}\lambda \right) - \frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i.$$

The parenthesized stationary terms cancel by Equation A2. The remaining perturbation dynamics are

$$\frac{d\delta g_i}{dt} = -\frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i. \quad (\text{A5})$$

To derive Equation A6, substitute the conductance and voltage decompositions from Equation A4 into the membrane equation, Equation 5:

$$C_E \frac{d(|v|_\lambda + \delta v_i)}{dt} = g_{L,E}(E_L - |v|_\lambda - \delta v_i) + (|g|_\lambda + \delta g_i)(E_e - |v|_\lambda - \delta v_i).$$

As above, the stationary mean has zero time derivative. Expanding and grouping the right-hand side gives

$$C_E \frac{d\delta v_i}{dt} = (g_{L,E}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda)) - (g_{L,E} + |g|_\lambda)\delta v_i + (E_e - |v|_\lambda)\delta g_i - \delta g_i\delta v_i.$$

The first parenthesized group is zero by Equation A3. The final product is second order in the small perturbations and is omitted by the local linear approximation. This leaves

$$C_E \frac{d\delta v_i}{dt} = -(g_{L,E} + |g|_\lambda)\delta v_i + (E_e - |v|_\lambda)\delta g_i. \quad (\text{A6})$$

Here C_E is excitatory-cell capacitance. All other symbols follow Equations A1–A4.

To derive Equation A7, take the Fourier transform of Equation A5. The derivative becomes $i\omega$ times the transformed signal:

$$i\omega\delta g_{i(\omega)} = -\frac{\delta g_{i(\omega)}}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_{i(\omega)}.$$

Move both conductance terms to the left, then divide by their common factor:

$$\left(i\omega + \frac{1}{\tau_{\text{AMPA}}}\right)\delta g_{i(\omega)} = w_{\text{probe}}\delta s_{i(\omega)},$$

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}}\delta s_{i(\omega)}. \quad (\text{A7})$$

Equation A8 follows in the same way from Equation A6. Fourier transformation first gives

$$i\omega C_E \delta v_{i(\omega)} = -(g_{\text{L,E}} + |g|_{\lambda})\delta v_{i(\omega)} + (E_e - |v|_{\lambda})\delta g_{i(\omega)}.$$

Collect the voltage terms on the left:

$$(i\omega C_E + g_{\text{L,E}} + |g|_{\lambda})\delta v_{i(\omega)} = (E_e - |v|_{\lambda})\delta g_{i(\omega)}.$$

Dividing by the coefficient of $\delta v_{i(\omega)}$ gives

$$\delta v_{i(\omega)} = \frac{E_e - |v|_{\lambda}}{i\omega C_E + g_{\text{L,E}} + |g|_{\lambda}}\delta g_{i(\omega)}. \quad (\text{A8})$$

Here ω is angular frequency, i is the imaginary unit, and each argument ω denotes a Fourier-domain signal.

Equations A5–A8 form a **linear time-invariant (LTI)** system after linearization about the operating point λ . Because the system has been linearized about the operating point, it is locally linear and time-invariant, so the standard transfer-function formalism applies. The canonical frequency-domain input–output relationship is

$$\delta v_{i(\omega)} = G_{\lambda(\omega)}\delta s_{i(\omega)}.$$

Here $G_{\lambda}(\omega)$ is the local transfer function relating input spike-train perturbations to membrane-voltage perturbations. For non-zero input, the input–output relationship can be rearranged as

$$G_{\lambda(\omega)} = \frac{\delta v_{i(\omega)}}{\delta s_{i(\omega)}}.$$

Thus the ratio follows from the standard input–output relationship rather than serving as an independent definition. To derive $G_{\lambda}(\omega)$, start from Equation A7:

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}}\delta s_{i(\omega)}.$$

Substitute this expression for $\delta g_{i(\omega)}$ into Equation A8:

$$\delta v_{i(\omega)} = \frac{E_e - |v|_{\lambda}}{i\omega C_E + g_{\text{L,E}} + |g|_{\lambda}} \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}}\delta s_{i(\omega)} \right).$$

Reorder the scalar factors and factor out $\delta s_{i(\omega)}$:

$$\delta v_{i(\omega)} = \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_{\lambda}}{i\omega C_E + g_{L,E} + |g|_{\lambda}} \right) \delta s_{i(\omega)}.$$

Comparing this result with the canonical LTI relationship identifies the coefficient multiplying $\delta s_i(\omega)$ as the transfer function:

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - |v|_{\lambda}}{i\omega C_E + g_{L,E} + |g|_{\lambda}}. \quad (\text{A9})$$

Equation A9 is therefore the synaptic filter multiplied by the membrane filter, not an arbitrarily chosen ratio. Its λ -dependence captures the change in membrane gain and effective time constant with mean conductance.

A.3 Finite-window averaging

Start from the baseline-subtracted Step 1 feature in Equation 7:

$$z_i = \frac{1}{T} \int_0^T (v_{i(t)} - E_L) dt.$$

Within the stationary linearized model used in this appendix, substitute the voltage decomposition from Equation A4:

$$z_i = \frac{1}{T} \int_0^T (|v|_{\lambda} + \delta v_{i(t)} - E_L) dt.$$

The operating-point voltage and leak reversal potential are constant in time, so their integral is their difference multiplied by T. Dividing by T gives

$$z_i = |v|_{\lambda} - E_L + \frac{1}{T} \int_0^T \delta v_{i(t)} dt.$$

Define the operating-point feature and its perturbation by

$$|z|_{\lambda} = |v|_{\lambda} - E_L, \quad \delta z_i = z_i - |z|_{\lambda}.$$

Subtracting the operating-point feature from the expanded expression for z_i cancels the constant term and leaves

$$\delta z_i = \frac{1}{T} \int_0^T \delta v_{i(t)} dt. \quad (\text{A10})$$

Thus δz_i is the fluctuation around the operating-point feature, obtained by averaging the voltage perturbation over the 200 ms presentation duration T. To derive the frequency response, represent the averaging operation as a causal rectangular kernel $a_T(u)$, where

$$a_{T(u)} = \frac{1}{T}, \quad 0 \leq u \leq T,$$

and $a_T(u)$ is zero outside that interval. Applying this kernel as a moving-average filter gives

$$\delta z_{i(t)} = \int_{-\infty}^{\infty} a_{T(u)} \delta v_{i(t-u)} du = \frac{1}{T} \int_0^T \delta v_{i(t-u)} du.$$

At the end of the presentation, set t equal to T . To put the resulting integral into the same form as Equation A10, introduce the new variable

$$t' = T - u, \quad dt' = -du.$$

When u is zero, t' is T ; when u is T , t' is zero. The substitution therefore reverses the integration limits. Its minus sign reverses them back:

$$\delta z_{i(T)} = \frac{1}{T} \int_0^T \delta v_{i(T-u)} du = -\frac{1}{T} \int_T^0 \delta v_{i(t')} dt' = \frac{1}{T} \int_0^T \delta v_{i(t')} dt'.$$

The name of an integration variable has no effect on the value, so replacing t' with t recovers Equation A10 exactly.

For any time-domain function $f(u)$, use the Fourier-transform convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(u) \exp(-i\omega u) du.$$

Here $\hat{f}(\omega)$ is the Fourier transform of $f(u)$. Applying this definition with $f(u)$ equal to the rectangular kernel $a_T(u)$, and naming the result $A_T(\omega)$, gives

$$A_{T(\omega)} = \int_{-\infty}^{\infty} a_{T(u)} \exp(-i\omega u) du = \frac{1}{T} \int_0^T \exp(-i\omega u) du.$$

Evaluating the integral and rearranging gives

$$A_{T(\omega)} = \frac{\exp(-i\omega T) - 1}{-i\omega T} = \frac{1 - \exp(-i\omega T)}{i\omega T}.$$

At zero frequency, the continuous limit of $A_T(\omega)$ is one, as expected: averaging does not change a constant signal. By the convolution theorem, the averaging filter multiplies the voltage spectrum:

$$\delta z_{i(\omega)} = A_{T(\omega)} \delta v_{i(\omega)}.$$

Substitute the input-voltage relation from Equation A9:

$$\delta z_{i(\omega)} = A_{T(\omega)} G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Define the complete local input-feature response by the canonical LTI relationship

$$\delta z_{i(\omega)} = H_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Comparing the coefficients multiplying $\delta s_i(\omega)$ identifies $H_\lambda(\omega)$. The two components of Equation A11 are therefore

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}, \quad H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (\text{A11})$$

Here $A_T(\omega)$ is the averaging-window response, $\exp$ is the exponential function, and $H_\lambda(\omega)$ is the complete response from input spike-train perturbation to averaged-voltage perturbation.

For the Bode comparison, the zero-frequency-normalized magnitude of the complete input-feature response is

$$B_{H(\omega)} = 20 \log_{10} \left(\frac{|H_{\lambda(\omega)}|}{|H_{\lambda(0)}|} \right). \quad (\text{A12})$$

Here B_H is magnitude in decibels; $|\cdot|$ is complex magnitude; and $\log_{10}$ is the base-ten logarithm. Only H_λ is plotted because it includes the averaging window and is the response used to predict the ANN feature variance.

A.4 Poisson-noise variance

A centred, unit-amplitude Poisson point process has a flat two-sided input power spectral density:

$$S_{\text{in}(\omega)} = \lambda. \quad (\text{A13})$$

Here $S_{\text{in}}(\omega)$ is input power per unit angular frequency and λ is mean spike rate. Passing this noise through Equation A11 gives

$$S_{z(\omega)} = |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)}. \quad (\text{A14})$$

Here $S_z(\omega)$ is predicted feature-noise power and $|H_\lambda(\omega)|^2$ is the transmitted fraction of input-noise power. Under the two-sided angular-frequency convention, total predicted variance is

$$\text{Var}_{\text{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H_{\lambda(\omega)}|^2 S_{\text{in}(\omega)} d\omega. \quad (\text{A15})$$

Here $\text{Var}_{\text{linear}}(z)$ is predicted feature variance and π is the circle constant; the remaining symbols follow Equations A11–A14. This result is exact for the stated linear stationary model, but only approximate for the finite, discrete, conductance-dependent probe. It is retained as an analytical diagnostic rather than as the generator of ANN inputs.